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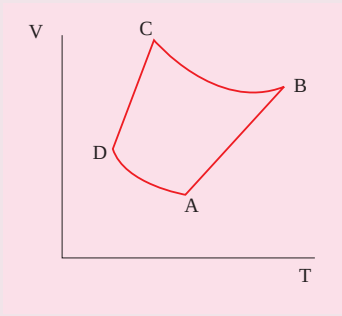
a) Process A

b)Process B

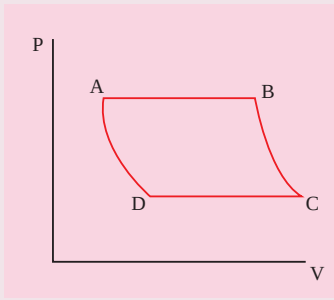
c) Process C

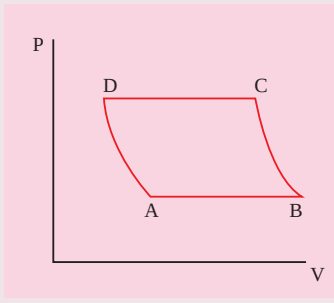
d) Equal work is done in Process A,B &C

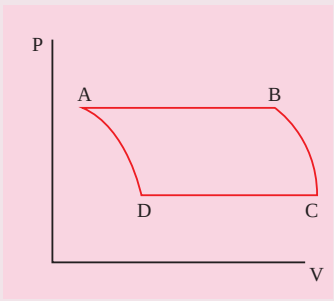
**10.** The V-T diagram of an ideal gas which goes through a reversible cycle $A \rightarrow B \rightarrow C \rightarrow D$ is shown below. (Processes $D \rightarrow A$ and $B \rightarrow C$ are adiabatic)

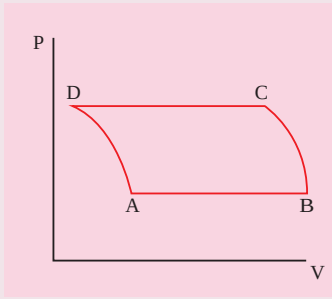



The corresponding PV diagram for the process is (all figures are schematic)

a)



b)



c)



d)



**11.** A distant star emits radiation with maximum intensity at 350 nm. The temperature of the star is

a) 8280 K b) 5000K

c) 7260 K d) 9044 K

**12.** Identify the state variables given here?

a) $Q$, $T$, $W$ b) $P$, $T$, $U$

c) $Q$, $W$ d) $P$, $T$, $Q$

**13.** In an isochoric process, we have

a) $W = 0$ b) $Q = 0$

c) $\Delta U = 0$ d) $\Delta T = 0$

**14.** The efficiency of a heat engine working between the freezing point and boiling point of water is

(NEET 2018)

a) 6.25% b)20%

c) 26.8% d)12.5%

**15.** An ideal refrigerator has a freezer at temperature −12°C. The coefficient of performance of the engine is 5. The temperature of the air (to which the heat ejected) is

a) 50°C (b)45.2°C

c) 40.2°C (d)37.5°C

**Answers:**

| | | | |
|---|---|---|---|
| 1) a | 2) c | 3) b | 4) b |
| 5) d | 6) a | 7) b | 8) c |
| 9) b | 10) b | 11) a | 12) b |
| 13) a | 14) b | 15) c | |